DO NOW — RECALL

Networks, security and compression practice

Name: _____ Class: _____ Date: _____

1. What does DNS resolve?

2. What is a router's role?

3. What is lossy compression?

4. 11010011_2 AND 00001111_2 as byte and hex?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Domain name to an IP address.

2. Model answer

Forwards IP packets between networks.

3. Model answer

Discards information irreversibly.

4. Model answer

$00000011_2 = 03_{16}$.

By the end, you can…

• Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

• RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

• Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

Vocabulary — know these terms

DNS — Maps names to IP addresses.

TLS — Authenticated secure transport.

RLE — Run-length encoding.

digest — Output of hash function.

NEW LEARNING

Read, annotate and explain

Understand

Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

Apply

RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

Evaluate

Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Encode AAABBCC with run-length encoding.

Predict first

Count consecutive symbols, preserving the order of groups.

Explain and check

Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe DNS, router and TLS roles when accessing HTTPS website. [6]

Question 2

RLE-encode AAAAABBBCCDD and explain how decoding works. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate transmitting official exam results as a lossy image versus an authenticated, encrypted, losslessly stored file. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Hashes can be decrypted using the public key to recover the input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.